Software Requirements Specification (SRS)

Hospital Management System

Project	Hospital Management System
Version	1.0
Prepared for	Software Engineering Project
Date	14-11-2026
Status	Updated working copy – based on supplied repository and templates
Source baseline	5D-Hospital_Managment_System-main ZIP; SRS_SE_PROJECT.xlsx; Use Case Flow PDF

Revision History

Version	Date	Author	Change Summary
1.0	14-11-2026	Project Team	Reformatted and expanded SRS to the supplied SE template; requirements checked against repository source.

Approvals

Role	Name	Signature / Date
Project Guide / Coordinator	_____	_____
Team Representative	_____	_____

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1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements of the Hospital Management System. It describes patient-facing and doctor-facing capabilities, external interfaces, security requirements, acceptance criteria and traceability to the planned test cases.

1.2 Scope

The system provides patient registration and phone verification, patient login, doctor discovery, appointment slot lookup and booking, appointment cancellation, medical history viewing, doctor diagnosis entry, prescription management, report upload and download, prescribed medicine ordering with a demonstration payment flow, and bed availability/management.

The supplied implementation is a demonstration application. The backend currently stores most application records in in-memory arrays; a persistent database, production SMS delivery and a real payment gateway are not evidenced in this repository and are outside the implemented scope.

1.3 Audience

Project evaluators, instructors, developers, testers, maintainers and other stakeholders.

1.4 Definitions and Acronyms

Term	Definition
SRS	Software Requirements Specification
API	Application Programming Interface
JWT	JSON Web Token
OTP	One-Time Password
UI	User Interface
RBAC	Role-Based Access Control
UT / IT / ST	Unit / Integration / System Test

2. Overall Description

2.1 Product Perspective

The product is a browser-based web application with a React frontend and an Express backend. The frontend calls JSON REST-style endpoints under /api. The backend performs validation, authentication, business rules and in-memory record management. Uploaded report files are stored in a backend private-reports directory.

2.2 Major Product Functions

- Patient registration, OTP verification and login
- Doctor directory and appointment slot availability
- Appointment booking, listing and cancellation
- Patient medical history and report metadata
- Doctor diagnosis and prescription entry
- Medical report upload and authorized file download
- Prescribed medicine catalogue, order creation and demo payment
- Doctor bed availability, allocation and release

2.3 User Roles and Characteristics

Role	Characteristics / Main capabilities
Patient	Registers, verifies phone, logs in, books/cancels appointments, views own history/prescriptions/reports, orders prescribed medicines and views bills.
Doctor	Logs in through doctor credentials, views assigned appointments, accesses relevant patient history, records diagnoses and prescriptions, manages reports and bed allocation.
Hospital staff	The existing component diagram depicts a staff portal, but the supplied code does not evidence a separate staff authentication role or staff-specific API. Treat as a proposed stakeholder, not a verified implemented role.

2.4 Operating Environment

Frontend: React 19 with Vite 8. Backend: Node.js with Express 5. Browser-based client and local HTTP API at http://localhost:5000/api in the current frontend source. Runtime configuration is loaded from backend/.env.

2.5 Constraints and Assumptions

- The backend requires JWT_SECRET of at least 24 characters and doctor credentials supplied through environment variables.
- Patient registration validates name, email, a 10-digit phone number and a password of at least eight characters.
- OTP is generated and logged to the backend console for development; an SMS gateway is not configured in the supplied source.
- Application records are held in process memory and reset when the backend restarts; the code does not configure MySQL or MongoDB.
- The payment endpoint is explicitly a demo flow, not a production payment integration.
- Medical report uploads accept PDF, JPEG and PNG files up to 5 MB.

3. External Interface Requirements

3.1 User Interfaces

The React application provides patient registration/login/OTP views, appointment booking and appointment history, medical records, prescriptions, bills and medicine purchase views. Doctor-facing views provide login, assigned appointment details, diagnosis, report and prescription actions, and bed availability. Screenshots in the repository provide UI evidence for key patient workflows.

3.2 Hardware Interfaces

No dedicated medical device, kiosk, scanner or hospital hardware interface is implemented or specified in the supplied source. Standard client devices and network connectivity are assumed.

3.3 Software Interfaces

Interface	Description
Frontend-to-backend API	React frontend sends HTTP requests to Express endpoints under /api, generally using JSON and bearer-token authorization.
Authentication	bcryptjs password hashing and jsonwebtoken signed tokens; patient OTP is hashed with SHA-256 and checked using timingSafeEqual.
File handling	Multer accepts PDF/JPEG/PNG attachments and stores them in backend/private-reports.
Configuration	dotenv loads JWT_SECRET and doctor account settings from backend/.env.

3.4 Communications

The local frontend source uses HTTP to localhost:5000. Production TLS termination, domain configuration, API gateway, retry policy and external messaging/payment integrations are not configured in the supplied ZIP and must be addressed before deployment.

4. System Features and Functional Requirements

The following 15 requirements decompose the 10 functional requirement groups in the supplied SRS spreadsheet into verifiable behaviours. Existing FR-01 to FR-10 identifiers are retained in the traceability column.

Req ID	Feature	Requirement (shall)	Priority	Acceptance Criteria	Source
HMS-FR-001	Patient registration	The system shall allow a patient to register using name, email, 10-digit phone number and password.	High	Valid details create an unverified patient account; duplicate email/phone is rejected.	FR-01
HMS-FR-002	OTP verification	The system shall verify a registered patient's phone number using a six-digit OTP.	High	Correct unexpired OTP verifies the phone; invalid, expired or excessive attempts are rejected.	FR-02
HMS-FR-003	Patient login	The system shall authenticate verified patients using email and password.	High	Valid credentials issue an eight-hour JWT; invalid credentials or unverified phone are rejected.	FR-01
HMS-FR-004	Doctor directory	The system shall display the available doctors and their specialties.	High	GET /api/doctors returns the configured doctor list.	FR-04
HMS-FR-005	Doctor slot availability	The system shall display available appointment slots for a selected doctor and date.	High	Past dates, invalid dates and already-booked slots are not offered.	FR-04
HMS-FR-006	Appointment booking	The system shall allow an authenticated patient to book an available slot with a doctor.	High	A valid request creates a booked appointment associated with the authenticated patient.	FR-04
HMS-FR-007	Appointment listing	The system shall allow a patient to retrieve their own appointments.	High	The endpoint returns only appointments associated with the authenticated patient.	FR-04
HMS-FR-008	Appointment cancellation	The system shall allow a patient to cancel their	Medium	Only the owning patient can cancel an	FR-05

		own booked appointment.		appointment whose status is Booked.	
HMS-FR-009	Medical history	The system shall display the authenticated patient's medical history and report metadata.	High	Only the patient's own records and reports are returned by the patient medical-record endpoint.	FR-03
HMS-FR-010	Diagnosis entry	The system shall allow an authenticated doctor to save a diagnosis for an appointment assigned to that doctor.	High	Valid diagnosis text is stored with the appointment and patient identifiers.	FR-06
HMS-FR-011	Prescription management	The system shall allow a doctor to create validated prescriptions for an assigned patient's appointment.	High	Prescription items require a valid catalogue medicine, dosage, frequency and instructions; the prescription is associated with the patient.	FR-07
HMS-FR-012	Medical report management	The system shall allow a doctor to request or upload a medical report attachment for an assigned appointment.	High	Report metadata is stored; uploaded files are limited to PDF/JPEG/PNG and 5 MB.	FR-03/FR-09
HMS-FR-013	Medical file download	The system shall allow the owning patient or assigned doctor to download an available medical report.	High	Download requests are authorized by role and record ownership; unavailable files return an error.	FR-09
HMS-FR-014	Prescribed medicine purchase	The system shall allow a patient to view and order medicines present in their prescriptions.	Medium	Only prescribed medicines can be ordered; quantities and server-side prices are validated.	FR-08
HMS-FR-015	Bed availability and management	The system shall display bed availability to authenticated doctors and allow authorized doctors to allocate or release beds.	Medium	Bed list includes availability summary; allocation/release rejects invalid state transitions.	FR-10

5. Non-Functional Requirements

Req ID	Category	Requirement	Priority	Acceptance Criteria
HMS-NFR-001	Security	The system shall protect patient and medical information through authentication, role checks and ownership validation.	High	Unauthenticated requests are rejected; unauthorized cross-patient/role access returns 401/403.
HMS-NFR-002	Data protection	Sensitive credentials shall not be stored as plaintext passwords; password verification shall use a password-hashing function.	High	Passwords are stored using bcryptjs hashing; source review confirms no plaintext password field.
HMS-NFR-003	Usability	The system shall provide a straightforward interface for patient registration, appointments, medical records and medicine workflows.	High	A user can navigate the represented patient and doctor workflows and receive validation/status messages.
HMS-NFR-004	Language support	The system should support regional language selection and display of localized interface content.	Medium	Acceptance requires language selection and localized content; this is not evidenced as implemented in the supplied source.
HMS-NFR-005	Availability and reliability	The system shall provide reliable access to essential services and preserve application records during normal operation.	High	Health endpoint responds with status ok; persistence across server restart is not met by the current in-memory data implementation.

5.1 Security Objectives

1. Confidentiality: patient identity, medical history, prescriptions and reports shall be accessible only to authenticated and authorized users.

2. Integrity and accountability: only permitted users shall create or change records; server-side validation and ownership checks shall prevent unauthorized operations.

5.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance / Verification
HMS-SR-001	The system shall authenticate users before granting access to protected application functions.	Authentication	High	Protected endpoints reject requests without a valid bearer token.
HMS-SR-002	The system shall enforce role-based access for patient and doctor functions.	Authorization	High	Doctor-only endpoints reject patient tokens; patient endpoints require patient authentication.
HMS-SR-003	The system shall restrict access to patient medical records to authorized users.	Confidentiality	High	A user cannot retrieve another patient's records.
HMS-SR-004	The system shall hash passwords before storing them.	Credential protection	High	bcryptjs hash is stored and bcrypt comparison is used at login.
HMS-SR-005	The system shall verify phone ownership through OTP before permitting patient login.	Identity verification	Medium	Unverified patients receive an authentication rejection.
HMS-SR-006	The system shall use signed, expiring authentication tokens for authenticated sessions.	Session security	High	JWT tokens include role and expire after eight hours.
HMS-SR-007	The system shall validate user-supplied values on the server before processing them.	Input validation	High	Invalid formats, missing fields, excessive lengths and invalid state transitions are rejected.
HMS-SR-008	The system shall restrict medical report uploads to permitted file types and file sizes.	File security	High	Only PDF, JPEG and PNG files up to 5 MB are accepted by the upload middleware.
HMS-SR-009	The system shall authorize medical file downloads based on patient ownership or assigned doctor relationship.	Access control	High	Unauthorized downloads return 403; missing files return 404.
HMS-SR-010	The system shall protect payment-related data and avoid storing payment credentials in plaintext.	Payment security	High	The current application exposes a demo payment endpoint; production payment-gateway security remains outside the demonstrated implementation.
HMS-SR-011	The system shall limit repeated OTP verification attempts.	Abuse prevention	Medium	OTP verification blocks further attempts after five incorrect submissions.
HMS-SR-012	The system shall keep authentication secrets in environment configuration rather than source code.	Secret management	High	JWT_SECRET and doctor credentials are read from backend/.env configuration.
HMS-SR-013	The system shall restrict prescription creation to the doctor assigned to the appointment.	Integrity	High	A doctor cannot create a prescription for an appointment assigned to another doctor.
HMS-SR-014	The system shall prevent patients from modifying or cancelling another patient's appointment.	Integrity	High	Appointment ownership is checked before cancellation.

6. Quality Attributes and Acceptance Tests

Quality Attribute	Acceptance Evidence
Security	Authentication, role checks, patient ownership checks, OTP attempt limits and private report download authorization are verified.
Functional correctness	Planned UT/IT/ST cases cover registration, appointments, medical records, prescriptions, payment, downloads and beds.
Usability	Patient and doctor workflows provide visible validation, loading, success and error feedback.
Reliability	Health endpoint and normal workflow checks; persistence requirement is currently limited by in-memory storage.
Maintainability	Frontend and backend are separated into independent packages; backend routes and domain operations are organized in server.js.

Acceptance exit criteria: all agreed high-priority requirements are verified; no unresolved critical defects remain; all test cases have recorded actual results; and the RTM identifies any untested or unsupported requirement. The supplied test-plan workbook has 100 planned cases with actual-result and test-result cells currently blank.

7. System Models and Use Cases

The existing Use Case Flow PDF defines the following ten use cases:

Use Case ID	Use Case	Primary Actor
UC-01	Patient Registration & Login	Patient
UC-02	OTP Verification	Patient
UC-03	View Medical History	Patient / Doctor
UC-04	Doctor Availability & Appointment Booking	Patient
UC-05	Appointment Cancellation	Patient
UC-06	Medical Diagnosis & Reports	Doctor
UC-07	Prescription Management	Doctor / Patient
UC-08	Online Payment & Medicine Purchase	Patient
UC-09	File Download	Patient
UC-10	Check Bed Availability	Doctor / Hospital staff

The repository includes Hospital_Management_System_Use_Case_Diagram_Clean.drawio and Hospital_Management_System_Use_Case_Flow.pdf. These are retained unchanged in the project Documents folder.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement	SRS Section	Module	Test Case References
HMS-FR-001 to 003	Patient registration, OTP and login	4.1	Authentication	UT-01–UT-04; IT-01–IT-03; ST-01–ST-03
HMS-FR-004 to 007	Doctor listing, slots and booking/listing	4.1	Appointment	UT-05–UT-09; IT-04–IT-06; ST-04–ST-06
HMS-FR-008	Appointment cancellation	4.1	Appointment	UT-17–UT-20; IT-13–IT-15; ST-13–ST-15
HMS-FR-009	Medical history	4.1	Medical Records	UT-21; IT-16; ST-16
HMS-FR-010	Diagnosis entry	4.1	Doctor / Diagnosis	UT-22–UT-24; IT-17–IT-18; ST-16–ST-18
HMS-FR-011	Prescription management	4.1	Prescription	UT-25–UT-28; IT-19–IT-21; ST-19–ST-21
HMS-FR-012 to 013	Reports and file download	4.1	File Management	UT-33–UT-36; IT-25–IT-27; ST-25–ST-27
HMS-FR-014	Medicine purchase and demo payment	4.1	Medicine / Payment	UT-29–UT-32; IT-22–IT-24; ST-22–ST-24
HMS-FR-015	Bed availability and management	4.1	Bed Management	UT-37–UT-40; IT-28–IT-30; ST-28–ST-30
HMS-NFR-001 to 002	Security and data protection	5	Authentication / API	Security cases; source review
HMS-NFR-003 to 004	Usability and language support	5	Frontend UI	UI acceptance review
HMS-NFR-005	Availability and reliability	5	Backend services	Health check; restart/persistence verification

Appendix A – Implementation Notes

The repository's existing SRS spreadsheet contains 10 functional requirement groups and 5 NFRs. The functional groups have been decomposed for clearer verification. The current backend is not database-backed; references in the legacy component diagram to a Hospital Database do not reflect the code in the supplied ZIP.